

Khyles Gibrian Ramos

Software Engineer · Web Developer

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GitHub: <https://github.com/NovaProtocol>

Portfolio: <portfolio.projectnova.download>



Summary

DOST Scholar and engineering student with a foundation in software programming, system automation, and network configuration. Experience in scripting, web infrastructure, and software-hardware integration through academic and personal projects. Adaptable and eager to learn, with a desire to build practical, real-world skills in system architecture and backend operations.

Career Objectives

Motivated individual seeking an entry-level technical role to apply hands-on experience in coding, circuit prototyping, and system simulations. Eager to contribute to a collaborative team while actively learning from experienced professionals and continuously expanding my multidisciplinary skill set.

Education

Southern Luzon State University 2021-2026

Bachelor of Science in Mechanical Engineering

DOST S&T Scholarship

Dean's Lister

Sacred Heart College Lucena 2019-2021

Science Technology Engineering & Mathematics (STEM) Strand

Academic Excellence Awardee - With Honours

Luis Palad Integrated High School 2015-2019

Junior High School

Certification

DOLE Accredited Safety Officer 2 (SO2) 2024

Basic Occupational Safety and Health (BOSH SO2) Training

Completed 40-Hour OSH Course

Work Experience

Southern Luzon State University Mar 2024 - Apr 2024

Physical Plant Facilities Department - PTP Program

Conducted routine inventory audits and applied asset tags to university property to improve tracking accuracy.
Inspected campus emergency lights and fire extinguishers to ensure proper working order and safety compliance.
Utilized AutoCAD to design and update detailed facility layouts and emergency fire exit plans.
Handled daily office operations, including documentation, message routing, and cross-department communications.

N/A - Work from Home Aug 2025 - Nov 2025

Freelance Software Developer

Developed a full-stack signature verification application using PyTorch, ResNet-18, and Flask to support academic thesis research.
Built a signature verification system that utilizes Triplet Loss to compare known reference signatures against unknown samples for authentication.
Developed a web-based user interface using Flask to allow users to upload signature samples and view real-time verification results.
Integrated PyTorch models into the backend to handle real-time image processing and model inference upon user request.
Created data preprocessing scripts to prepare, normalize, and format signature image datasets for model training.
Delivered functional code and technical documentation within project deadlines.

Projects

MELE Review

Tech Stack: FastAPI, Granian, SQLAlchemy, MySQL, Docker, Caddy, Cloudflare Tunnel

Self-hosted board exam reviewer with a self-maintained question bank from various sources, with answer feedback and flag-for-review

Notebook-style solution editor with a unit-aware solver that handles multi-step engineering equations in both metric and English conventions



Scan for my project portfolio

Water Billing System

Tech Stack: FastAPI, Granian, SQLAlchemy, MySQL, Docker, React Native, NFC, Xendit, Caddy, Cloudflare Tunnel

Designed a utility billing platform that keeps confidential customer and payment data isolated on internal Docker networks, with role-scoped API access so each user reaches only what they need

Built an offline-first meter reading app with tamper-proof NFC tags whose security keys derive on-device, so tag data never crosses the network

Automated billing from a 5-tier progressive tariff and reconciled 22 payment channels end-to-end

GateKeeper

Tech Stack: Flask, itsdangerous, SQLite, Docker, Cloudflare Tunnel

Single sign-on auth gate protecting a family of web apps with signed, tamper-evident cookies and forward-auth session verification

Admin panel to issue and revoke access codes, with a backup code so the owner is never locked out

SolveSpace

Tech Stack: Flask, MySQL, Docker, Bubblewrap, Cloudflare Tunnel

Practice sandbox that runs untrusted user code in isolated containers with hard memory, CPU, and process limits and no network access

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

Backend: FastAPI, Flask, Granian, SQLAlchemy, Jinja2

Infrastructure: Docker, Caddy, Cloudflare Tunnel, Linux

Databases: MySQL, SQLite

Mobile: React Native, Expo

Hardware: Arduino, ESP32, Raspberry Pi, NFC (NTAG215)

Mechanical: AutoCAD, OnShape, Fusion 360